

Botox Functionality Reference

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1 Introduction

Botox is a pure Rust compiler that turns Markdown documents into PDF files or HTML presentation slides. It embeds a native typesetting engine, so it does not require LaTeX, Node.js, Python, or Chromium to be installed on your machine.

This document describes all features implemented in Botox and tests their compilation.

2 Mathematics

Botox renders math using New Computer Modern fonts. You can write inline formulas like $E = mc^2$ or $\vec{F} = m\vec{a}$, as well as display equations.

2.1 Standard Display Equations

Display equations are enclosed in double dollar signs and can include Pandoc labels:

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi} \tag{1}$$

Equation Equation 1 is referenced using Pandoc cross-reference syntax. You can also write infinite series, fractions, and square roots:

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x, \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \tag{2}$$

2.2 Matrices and Cases

Parenthesized matrices (`\pmatrix`), bracketed matrices (`\bmatrix`), and piecewise functions (`\cases`) work out of the box:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (3)$$

Piecewise definitions use standard LaTeX `\cases` :

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \quad (4)$$

3 Diagrams and Visuals

Diagrams are compiled to vector SVG graphics via Kroki and cached locally in `~/.cache/botox/diagrams/` for fast repeated builds.

3.1 Inline Mermaid and PlantUML

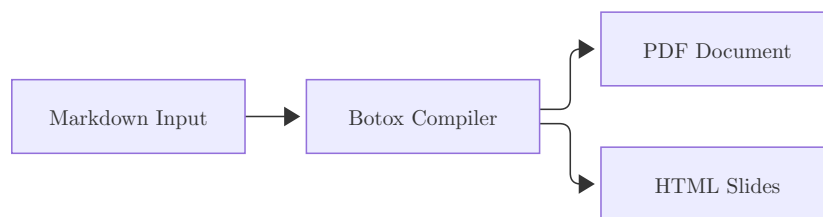


Figure 1: Compiler Pipeline

Figure Figure 1 shows the pipeline. PlantUML is also supported:

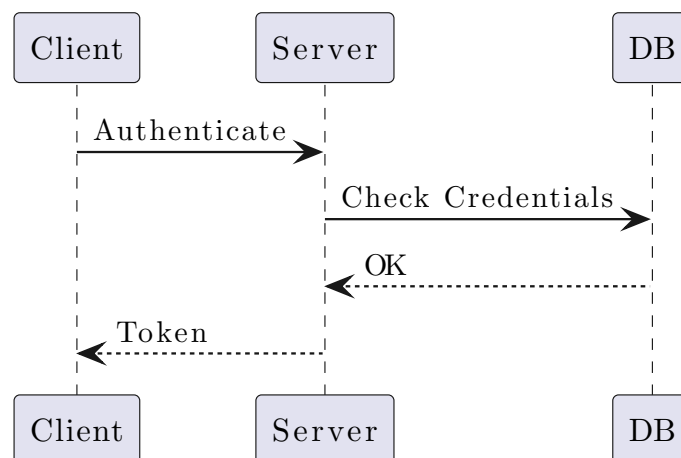


Figure 2: Authentication Sequence

3.2 UMLet Diagrams

Botox supports UMLet (`.uxf` format) diagrams both inline and via file references:



Figure 3: Inline UMLet Class Box

3.3 Diagram File Referencing

You can reference external diagram files directly without copying their contents:

- Through fence file attributes:

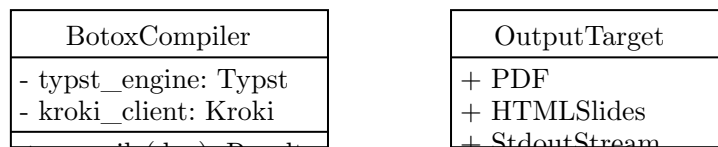


Figure 4: Referenced UMLet Architecture

- Through PlantUML include statements:

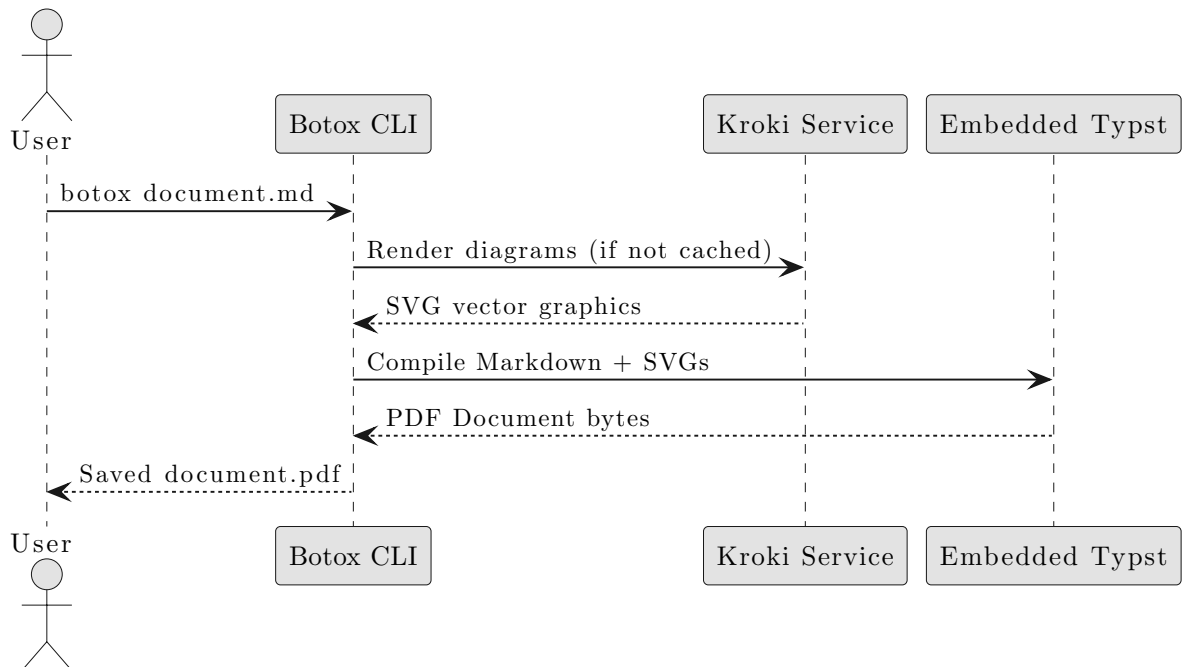


Figure 5: Included PlantUML Sequence

- Through standard Markdown image syntax:

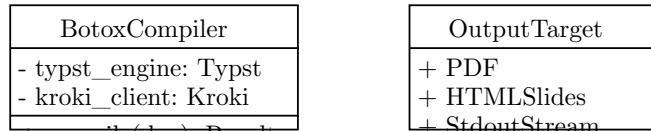


Figure 6: UMLet Architecture via Image Syntax

Figure Figure 4 and Figure Figure 6 reference external diagram files.

4 External Code File Referencing

Code blocks can include external source files directly using the `file` attribute:

```
// Sample source code referenced by documentation
pub struct BotoxConfig {
    pub theme: String,
    pub columns: usize,
    pub bibliography: bool,
}

impl BotoxConfig {
    pub fn default_document() -> Self {
        Self {
            theme: "academic".to_string(),
            columns: 1,
            bibliography: true,
        }
    }
}

fn main() {
    let _cfg = BotoxConfig::default_document();
}
```

5 Structured Tables

Tables support alignment, captions, and cross-references:

Target	File Extension	Engine	Offline Caching
Document	<code>.pdf</code>	Typst	Yes
Slides	<code>.html</code>	Custom HTML5	Yes
Diagrams	<code>.svg</code>	Kroki / mermaid.ink	Yes

Table 1: Botox compilation target summary.

As shown in Table Table 1, both document and presentation outputs are fully self-contained.

6 Callouts

Botox supports both GitHub-style callouts and Pandoc-style advisory divs.

Note

This is a GitHub-style note callout for background information.

Tip

Use `-w` or `--watch` to recompile automatically when files change.

Warning

Keep your diagrams valid so Kroki can generate vector output.

Important

Pandoc Style Div: This is an important announcement rendered with a colored frame.

7 Page Breaks and Date Macro

You can insert manual page breaks using `\newpage`, `\pagebreak`, or `<!-- pagebreak -->`.

Document generated on 09 October 2026.

8 References

External web links in the document are converted into IEEE numerical citations in the bibliography when bibliography mode is enabled:

- Read the [Typst Documentation](https://typst.app/docs/) [1] for typesetting details.
- Read the [Rust Language Guide](https://www.rust-lang.org) [2] for systems programming notes.
- Links to github like [Botox Repository](#) are excluded from the bibliography via frontmatter rules.

Works Cited

[1] “Typst Documentation”, [Online]. Available: <https://typst.app/docs/>.

[2] “Rust Language Guide”, [Online]. Available: <https://www.rust-lang.org>.